

EMG Stimulation Artefact Suppression

Benchmark Results Report [BATCH]

Thesis A&B · Ayush Ram · University of Sydney · n=11 subjects

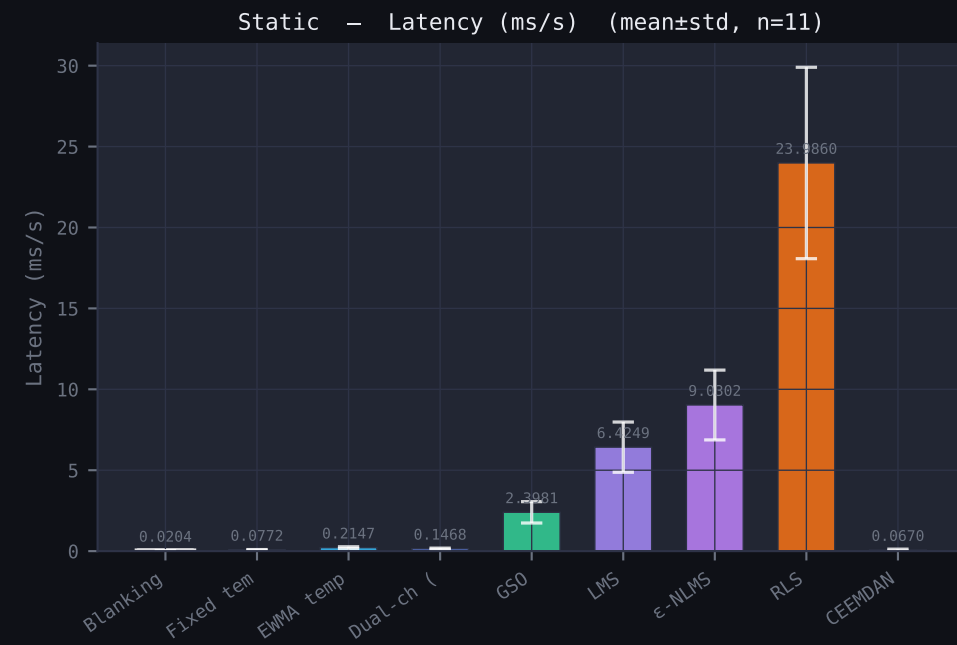
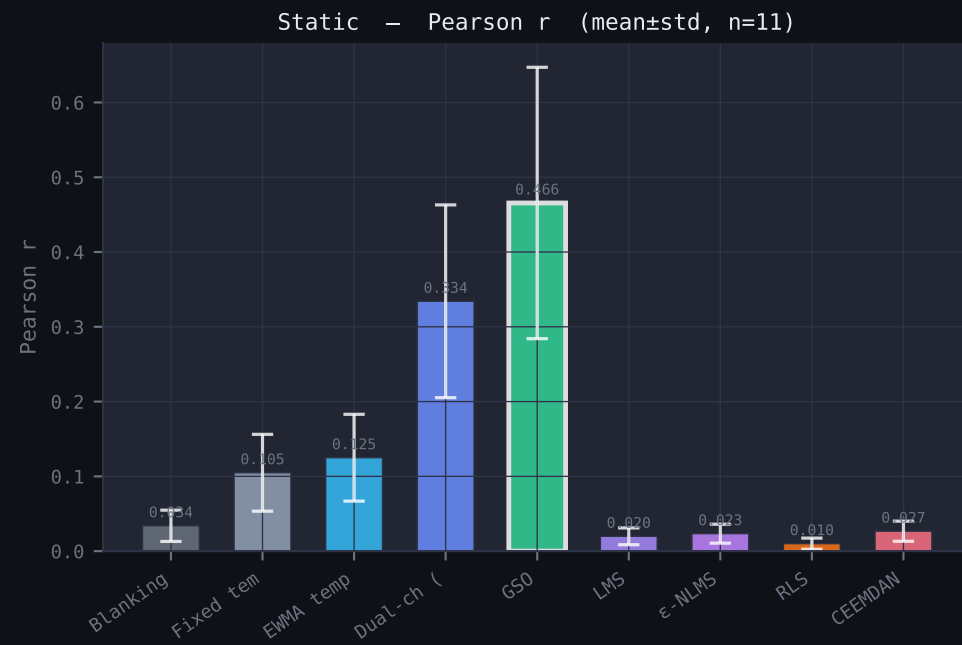
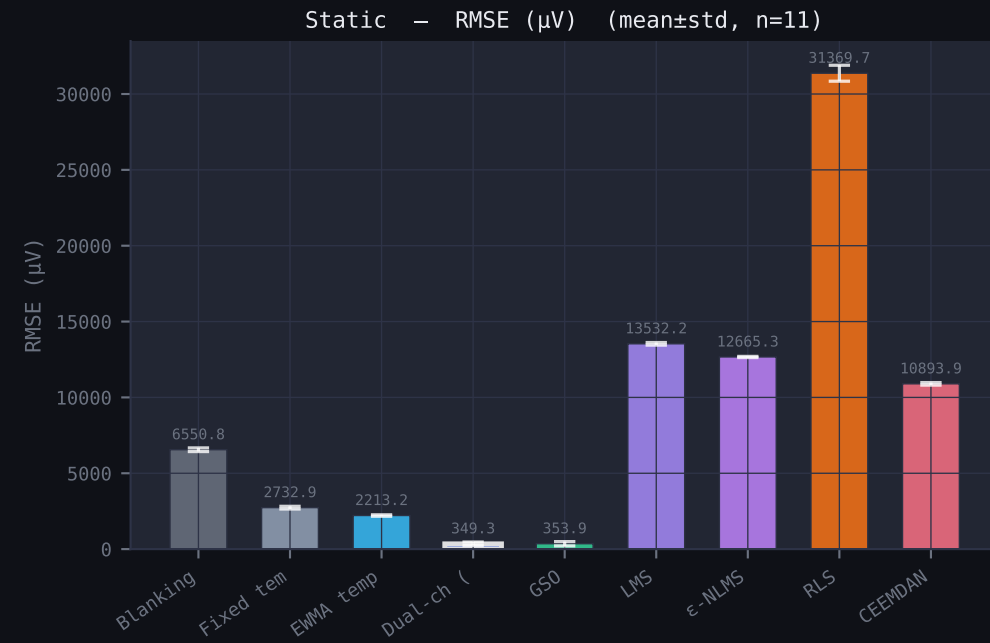
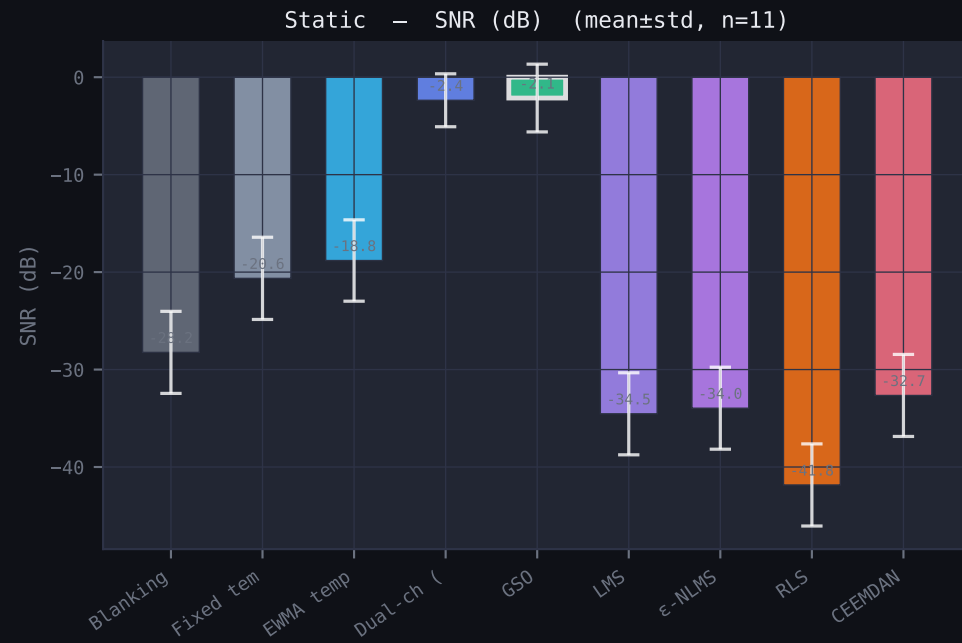
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Source: emg_batch_20260605_121247.xlsx | BATCH MODE — 11 subjects · 3 exercises

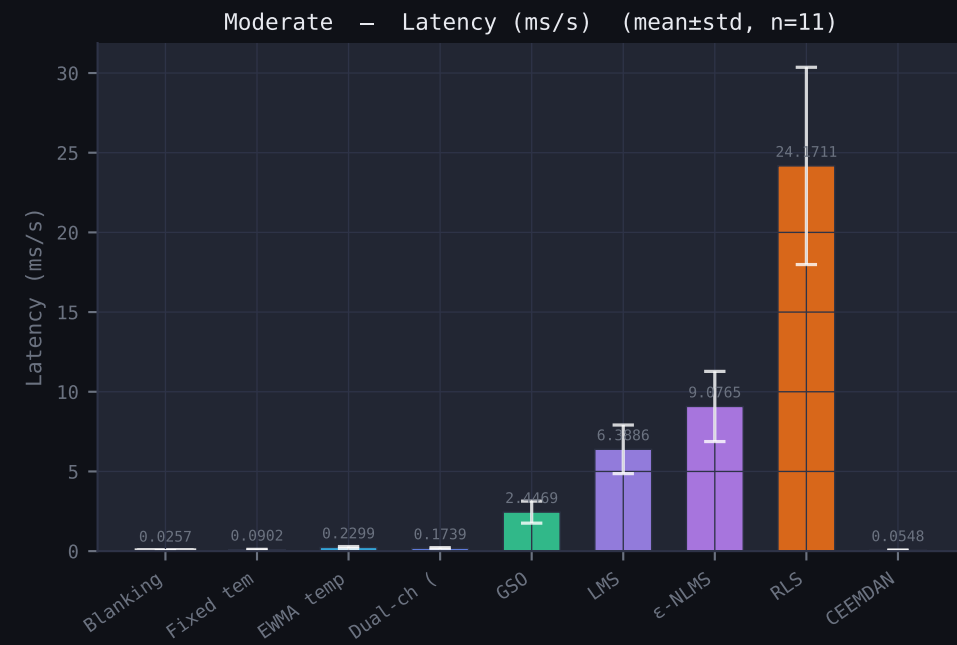
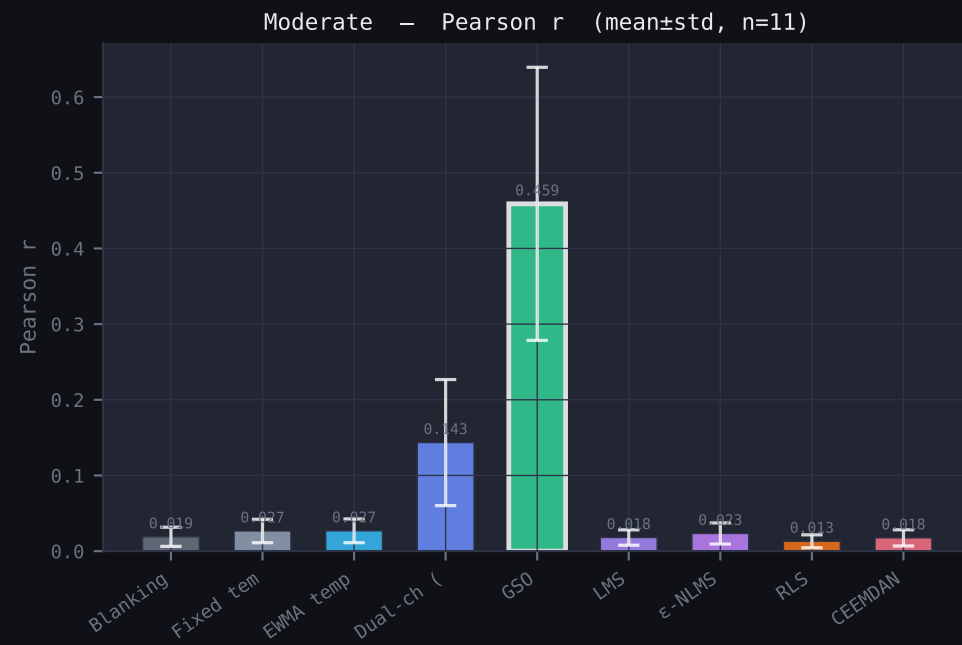
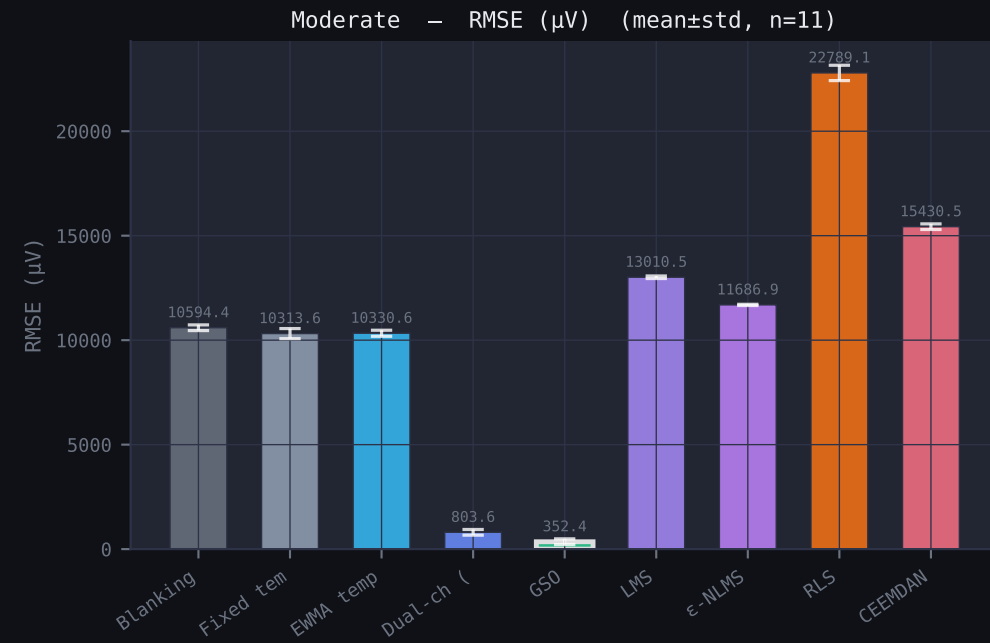
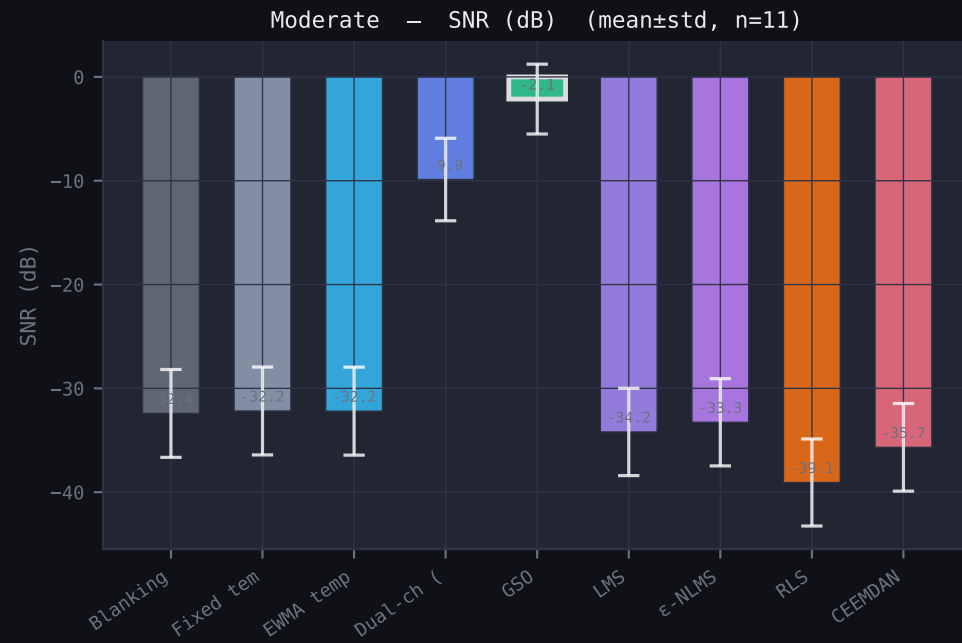
Algorithm colour key

 Blanking	 Fixed template	 EWMA template	 Dual-ch (DESTD)	 GS0
 LMS	 ϵ -NLMS	 RLS	 CEEMDAN	

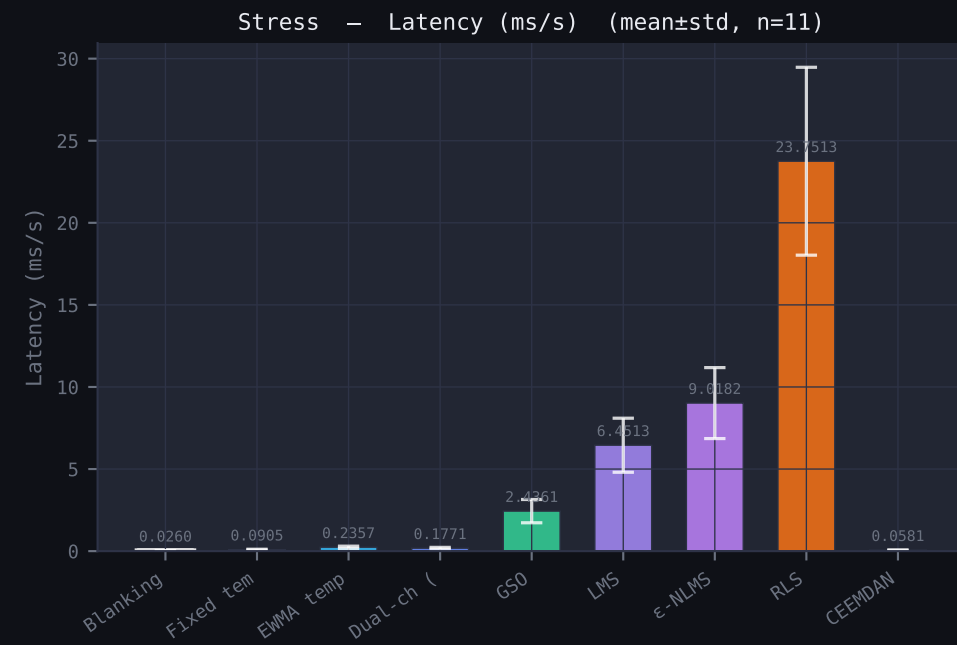
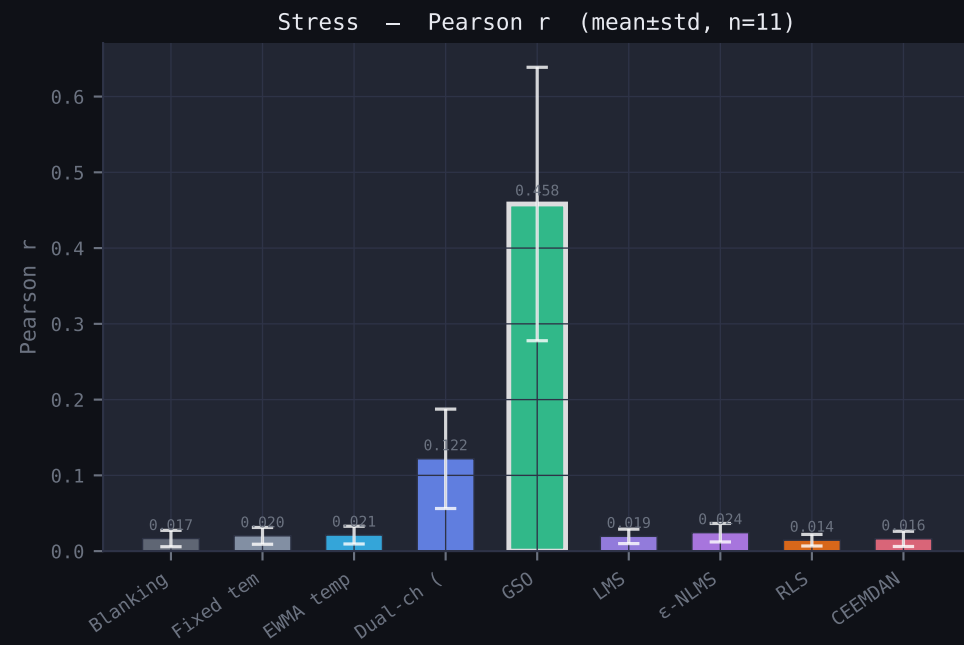
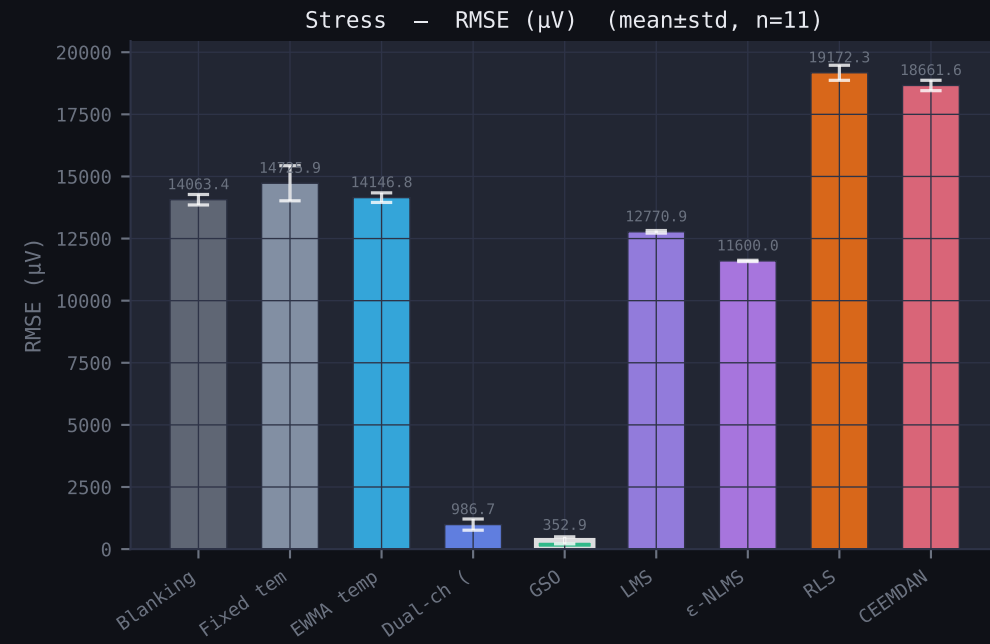
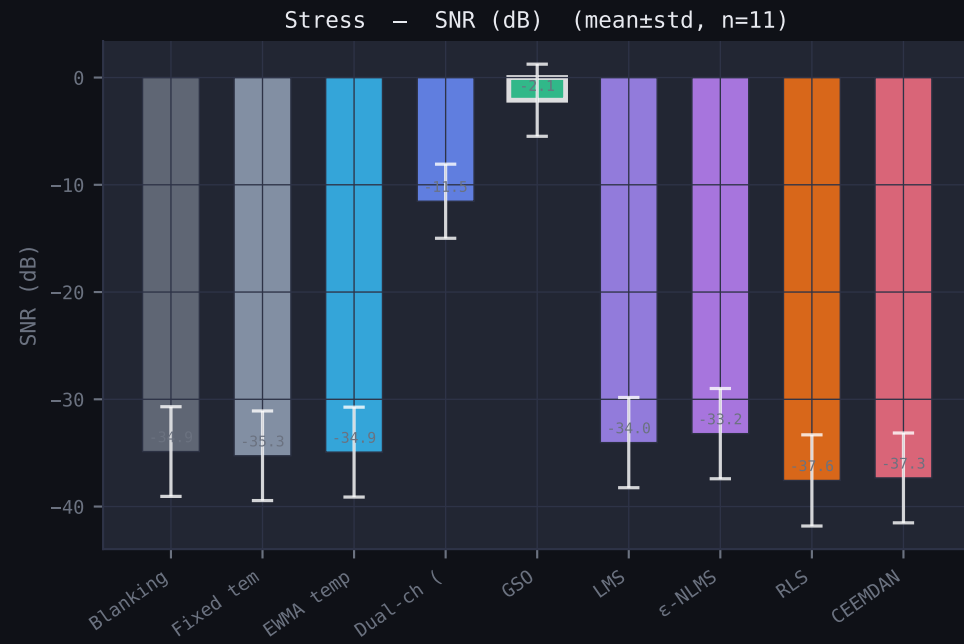
Algorithm performance – Static condition (mean±std, n=11)



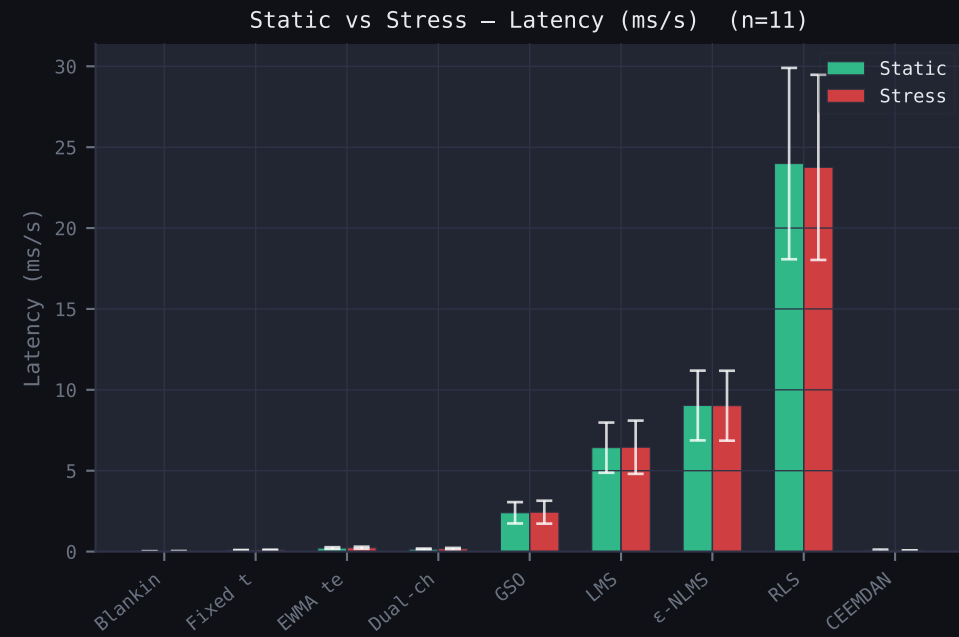
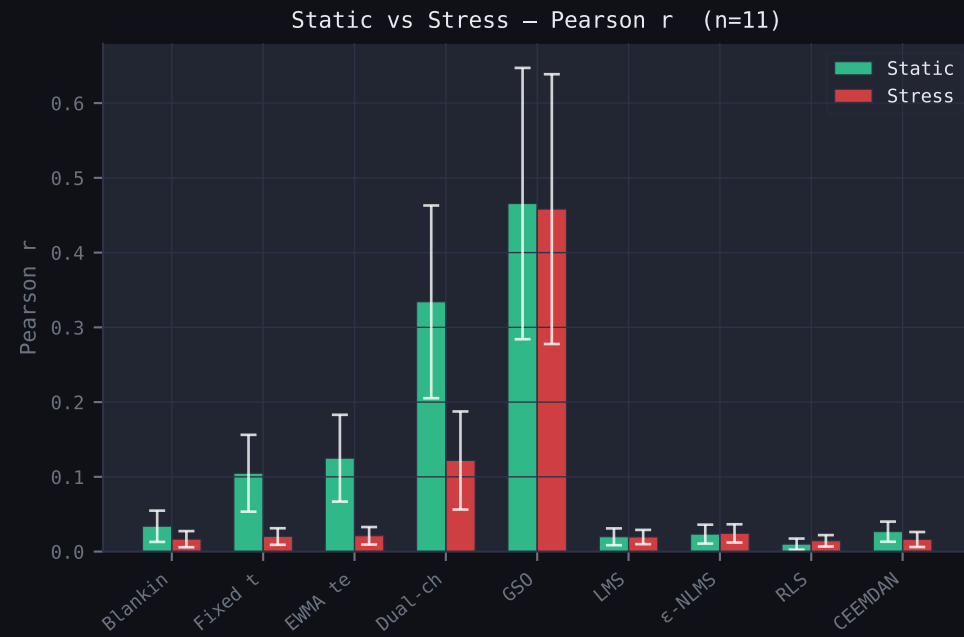
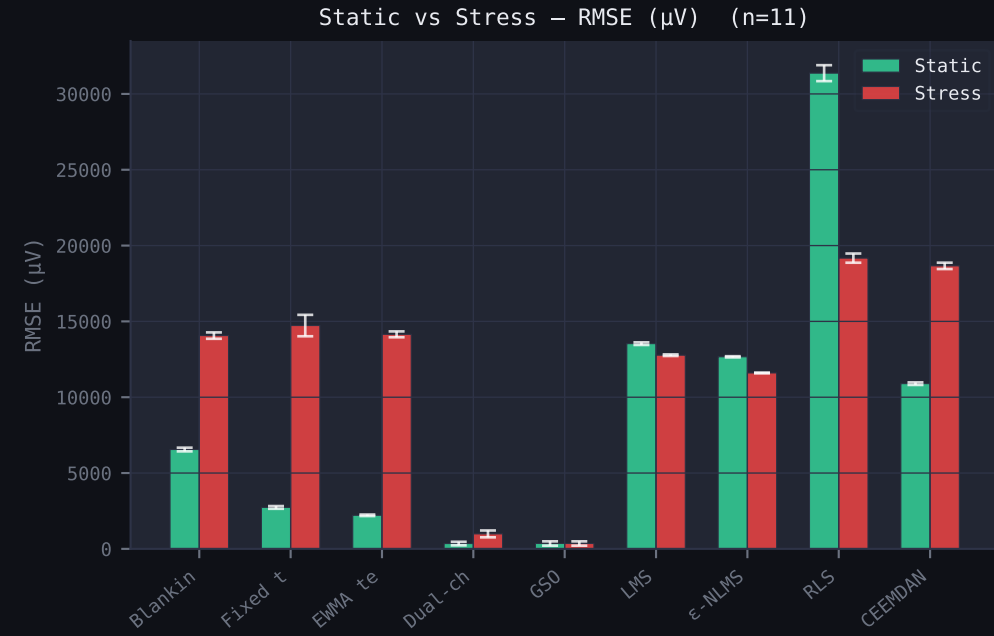
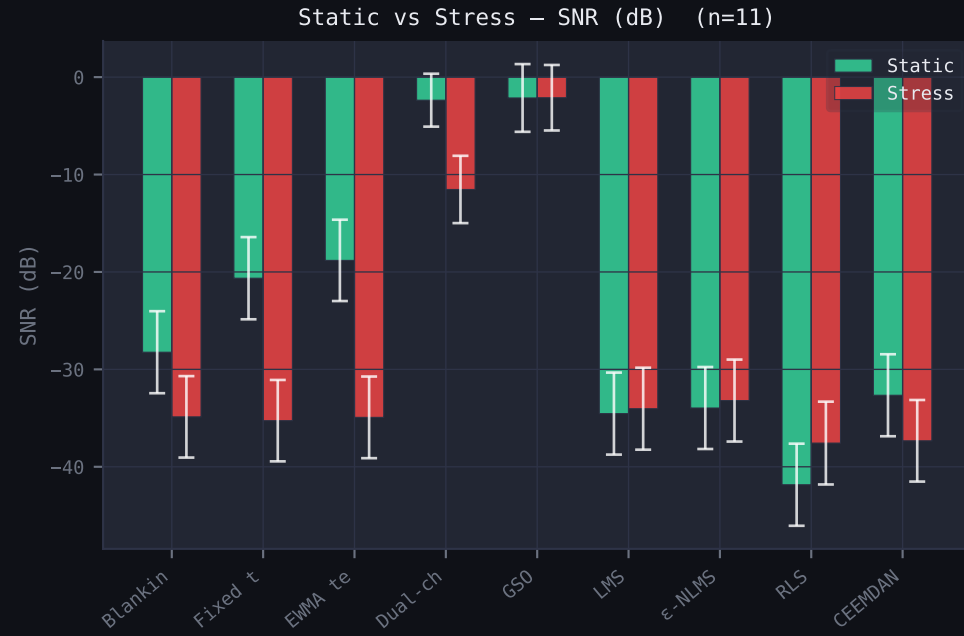
Algorithm performance – Moderate condition (mean±std, n=11)



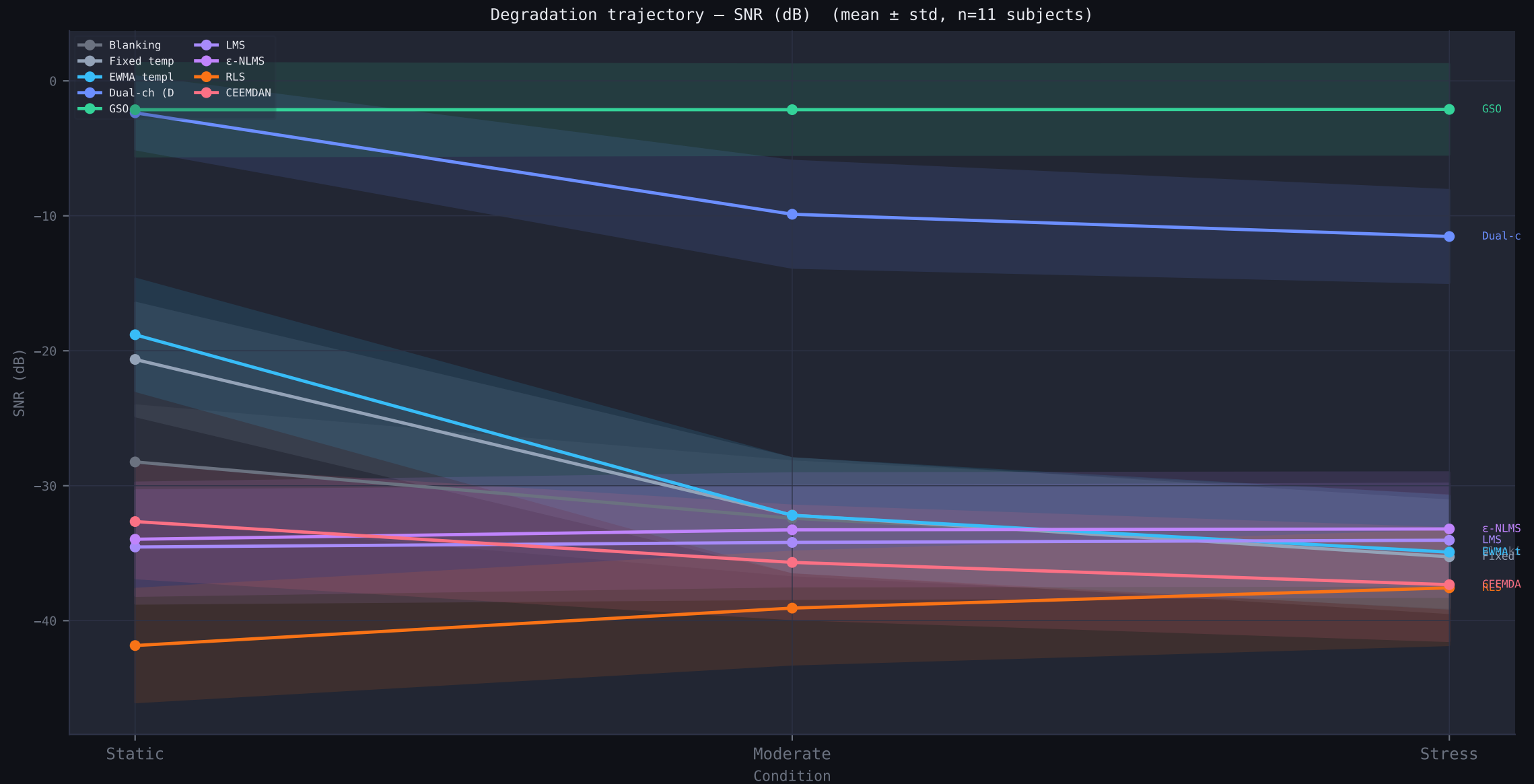
Algorithm performance – Stress condition (mean±std, n=11)



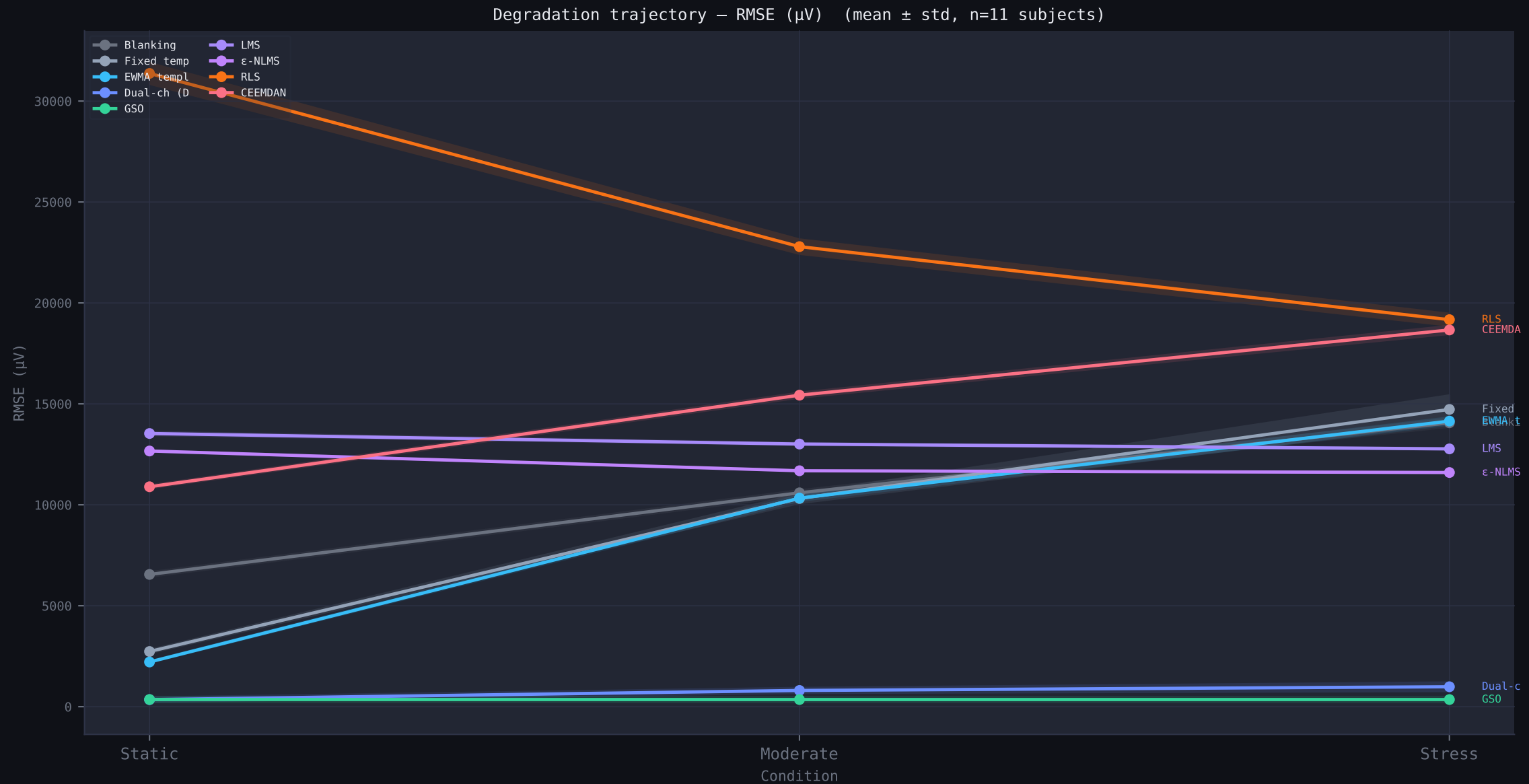
Static vs Stress – all metrics (mean±std, n=11)



Degradation trajectory – SNR (dB) (mean±std, n=11)

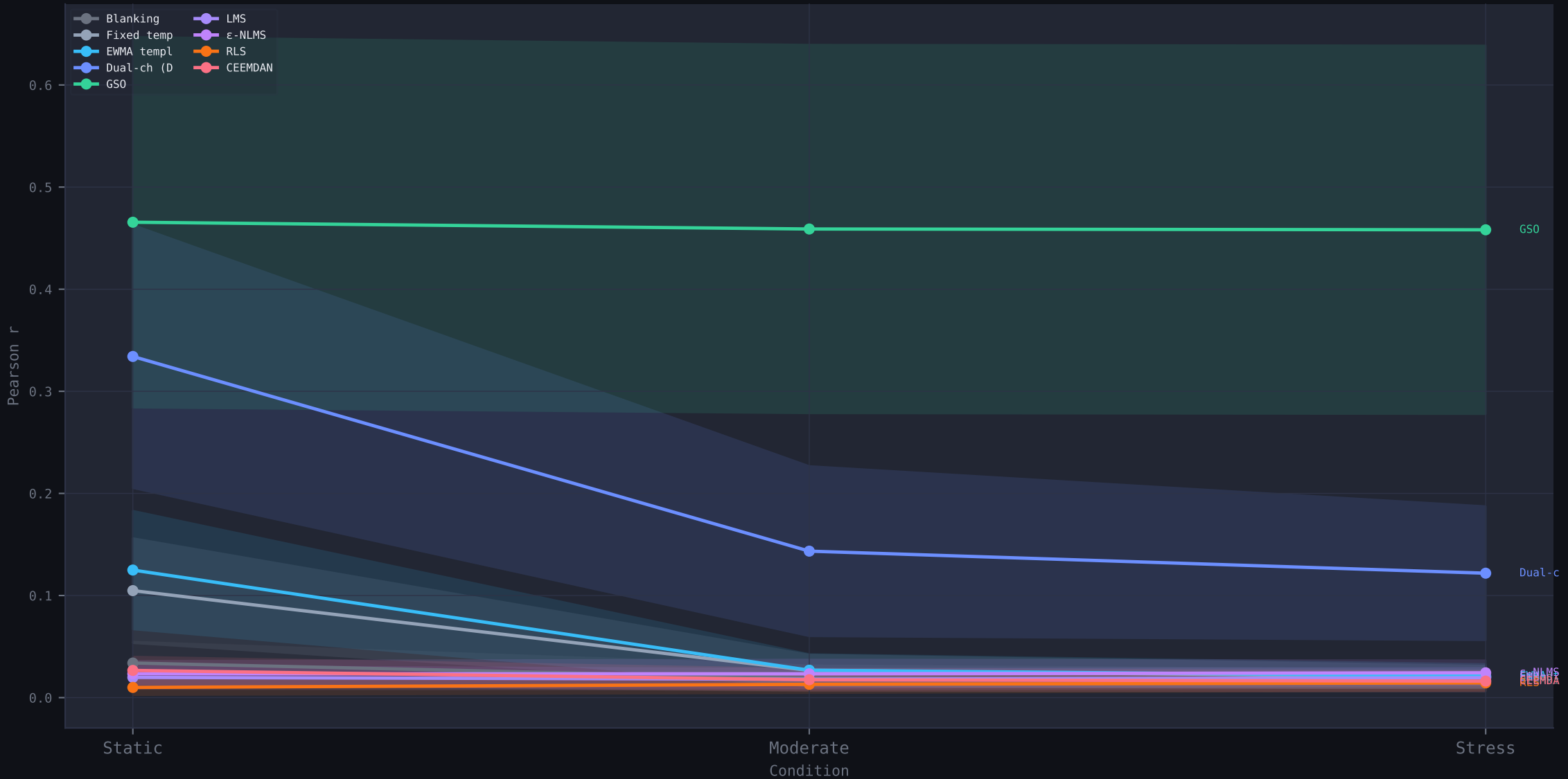


Degradation trajectory – RMSE (μV) (mean $\pm$ std, n=11)

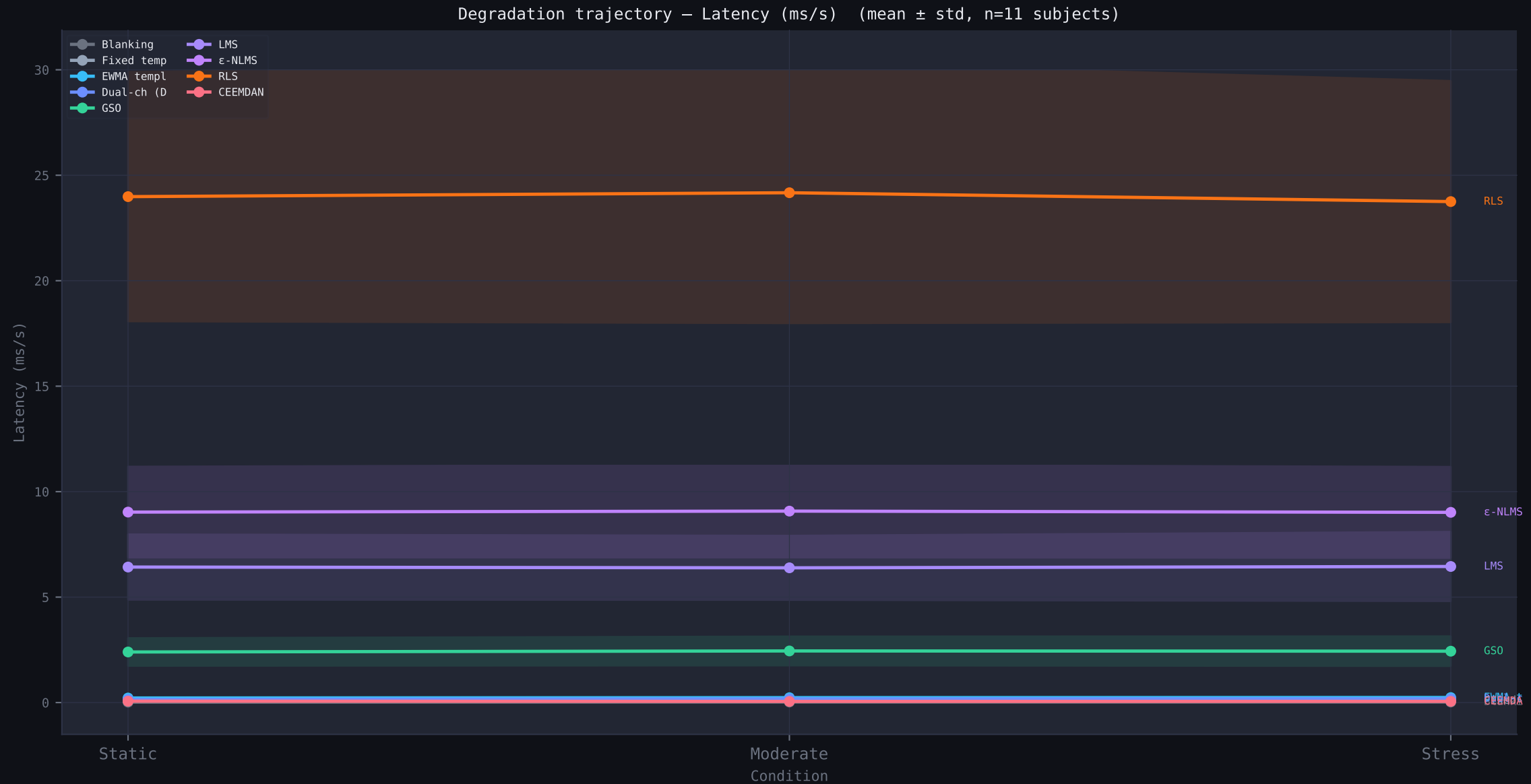


Degradation trajectory – Pearson r (mean±std, n=11)

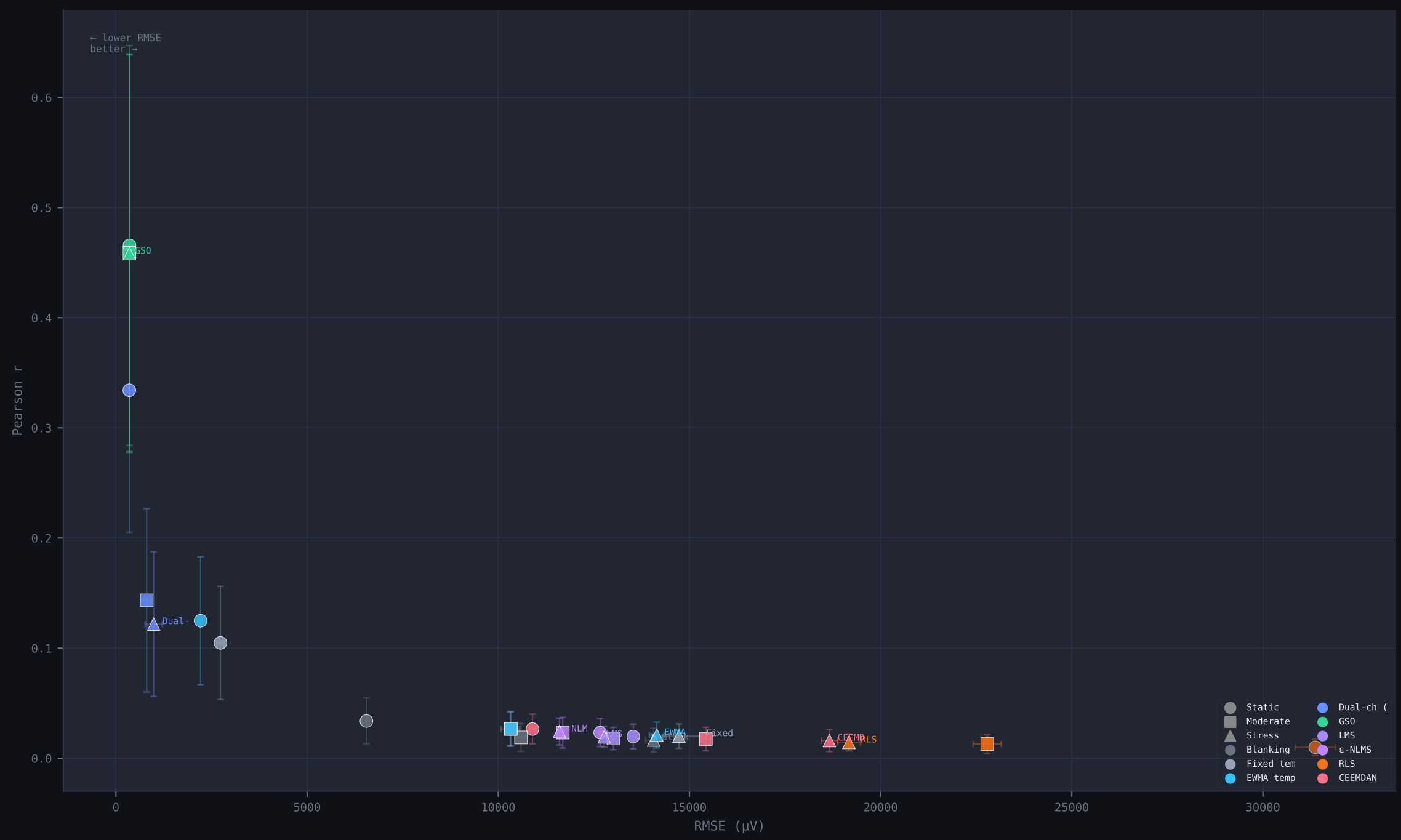
Degradation trajectory – Pearson r (mean ± std, n=11 subjects)



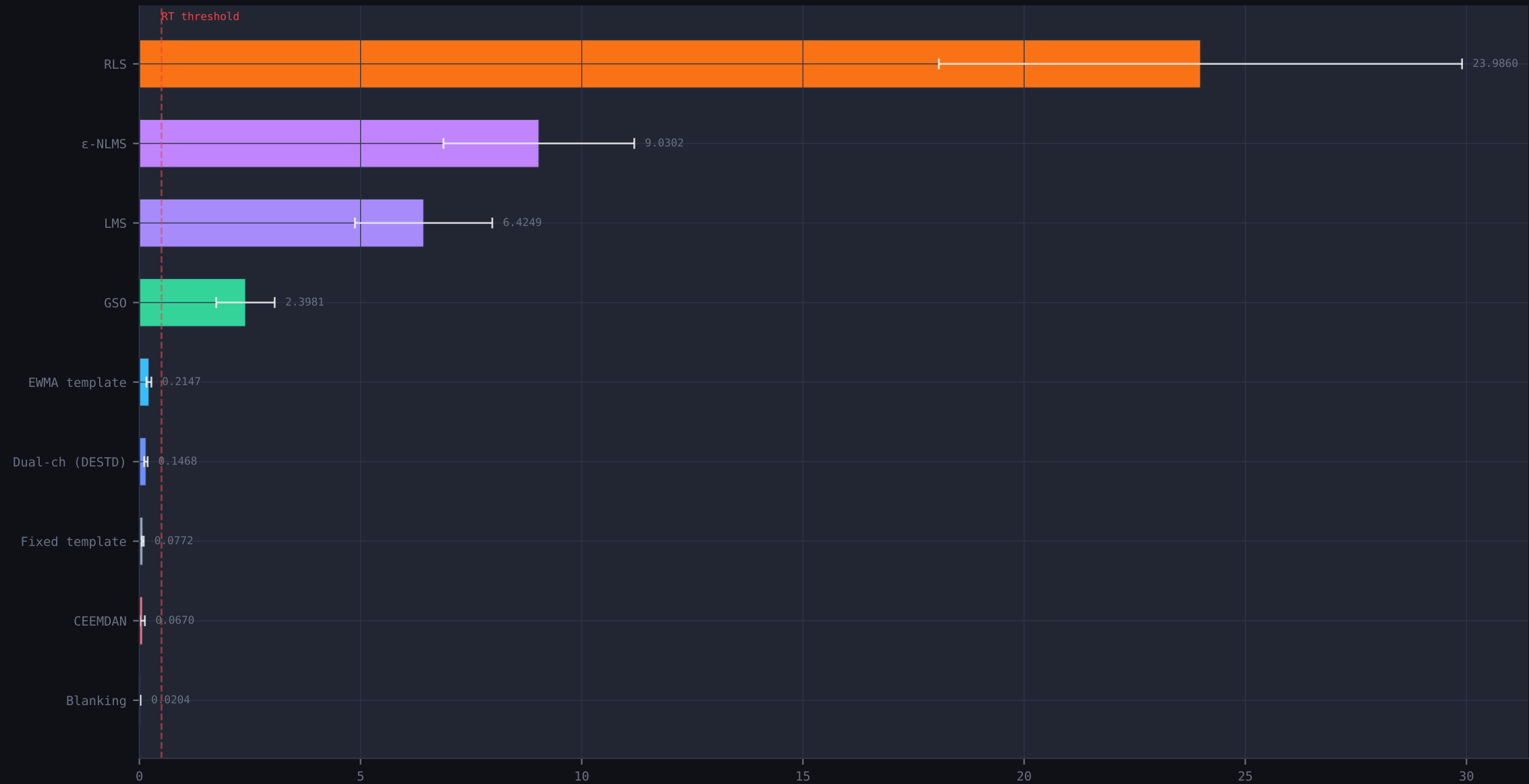
Degradation trajectory – Latency (ms/s) (mean±std, n=11)



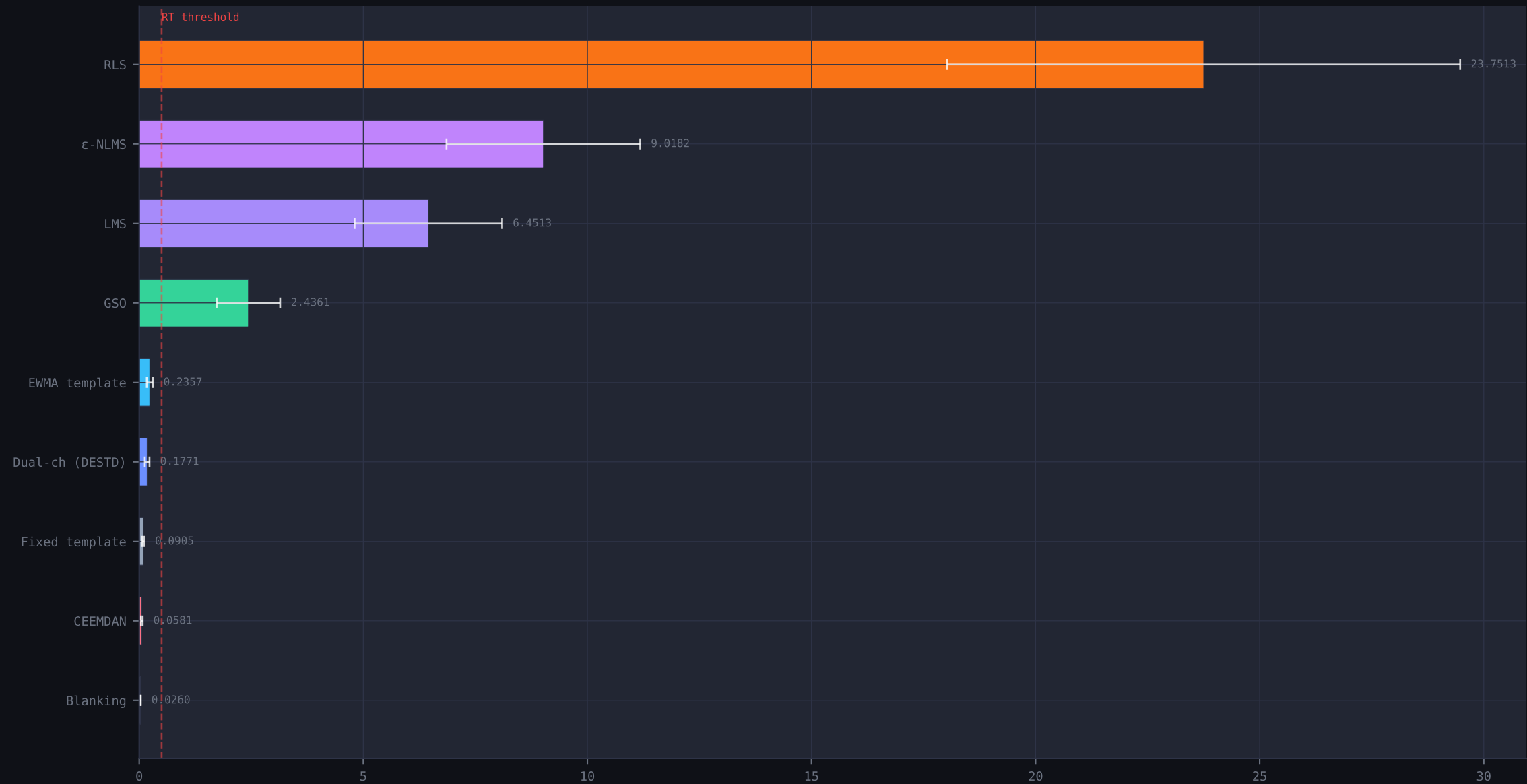
Quality trade-off space: RMSE × Pearson r



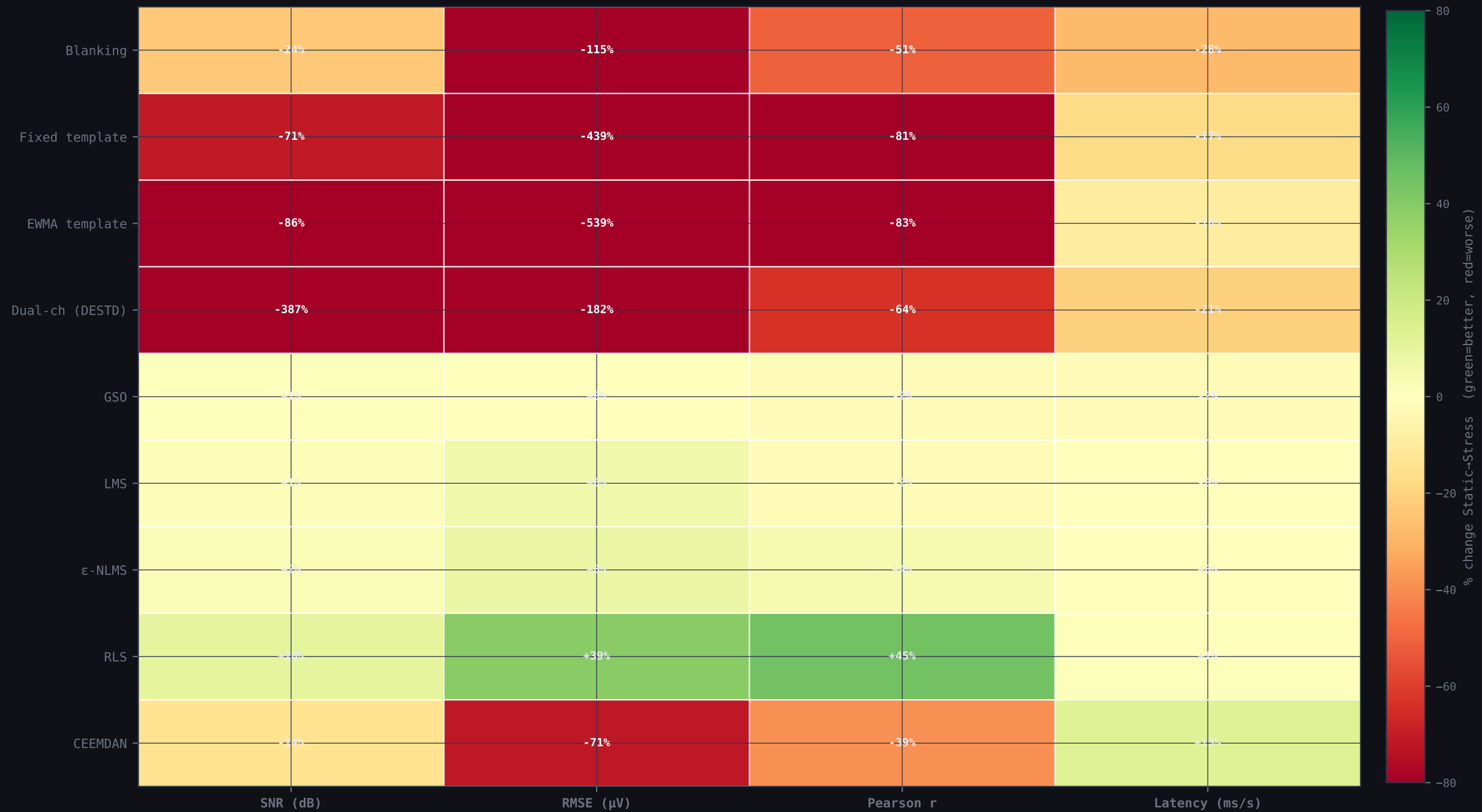
Latency ranking – Static



Latency ranking – Stress

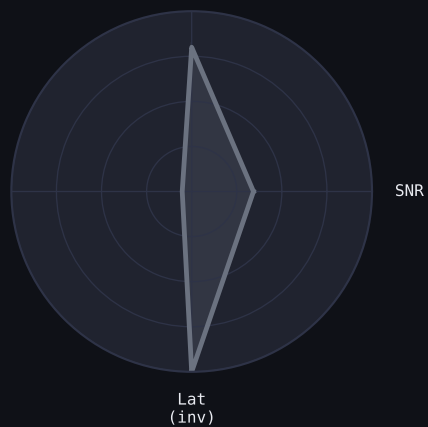


Robustness: % change Static → Stress



Multi-metric radar fingerprints – Static

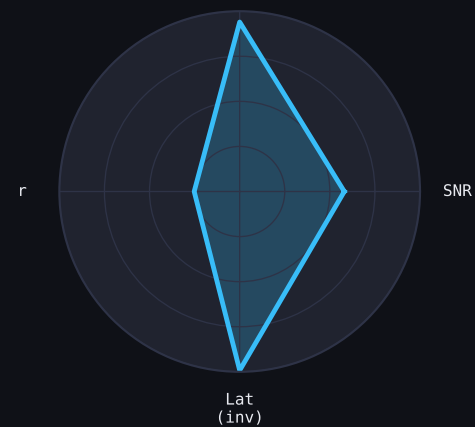
Blanking
RMSE
(inv)



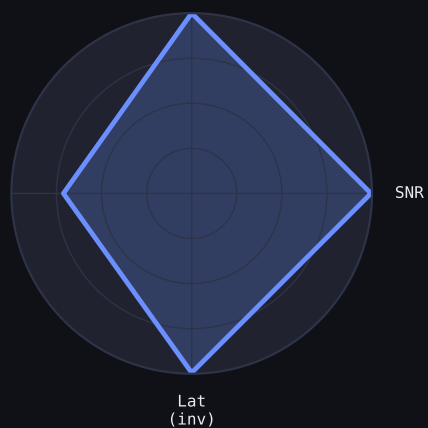
Fixed templ
RMSE
(inv)



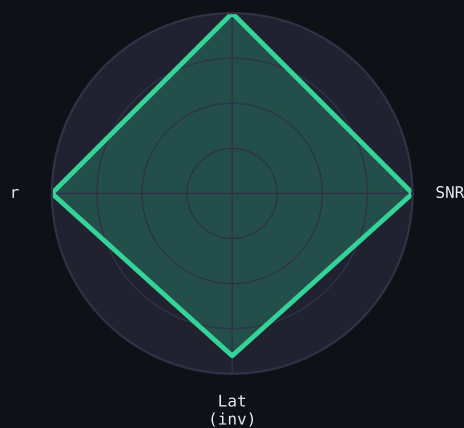
EWMA templa
RMSE
(inv)



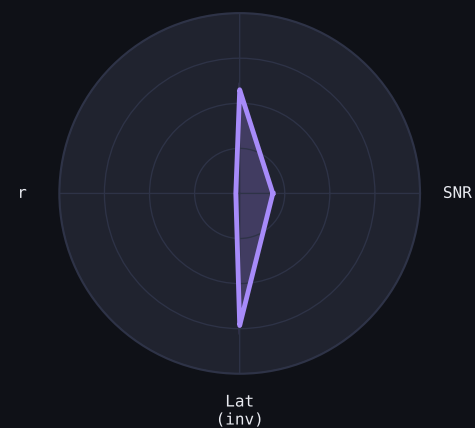
Dual-ch (DE
RMSE
(inv)



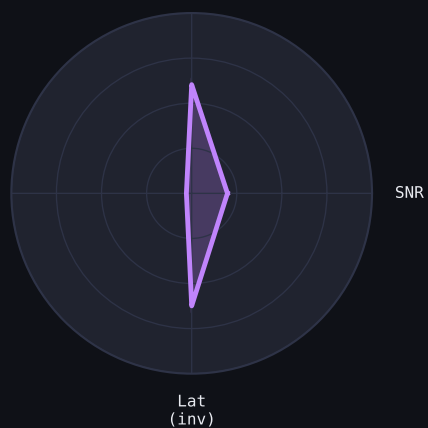
GS0
RMSE
(inv)



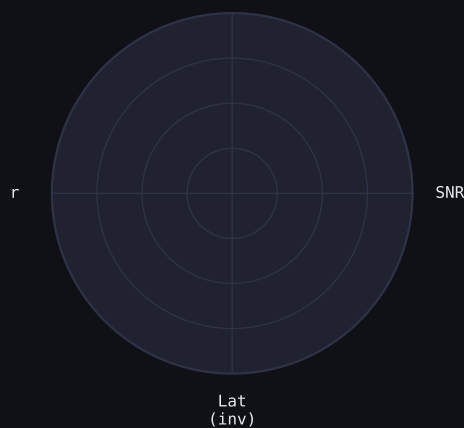
LMS
RMSE
(inv)



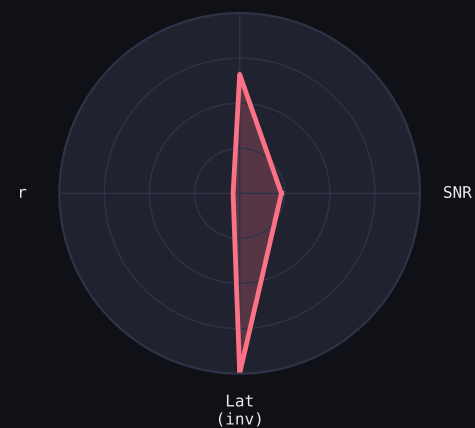
ϵ -NLMS
RMSE
(inv)



RLS
RMSE
(inv)

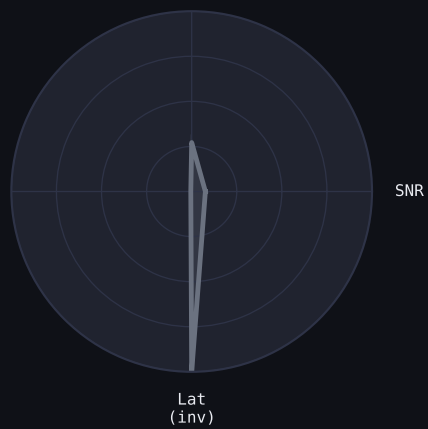


CEEMDAN
RMSE
(inv)



Multi-metric radar fingerprints – Stress

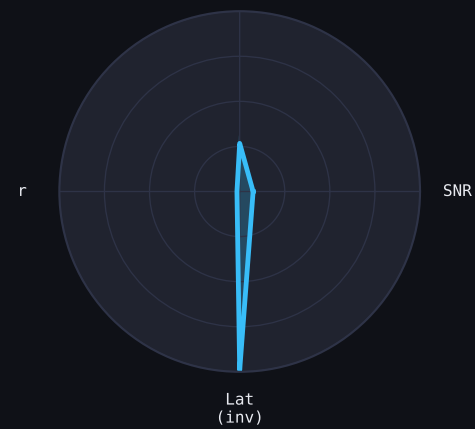
Blanking
RMSE
(inv)



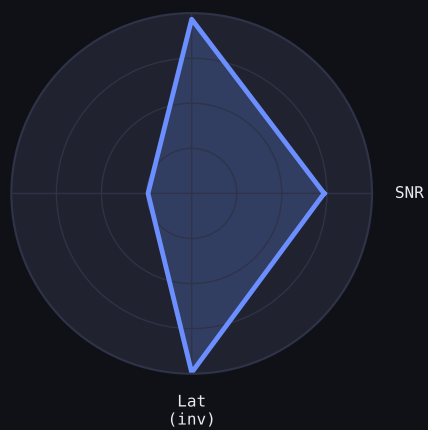
Fixed templ
RMSE
(inv)



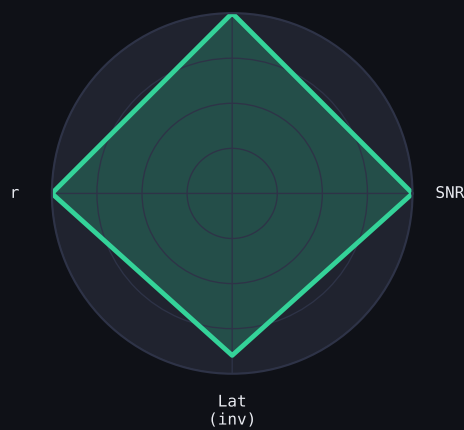
EWMA templa
RMSE
(inv)



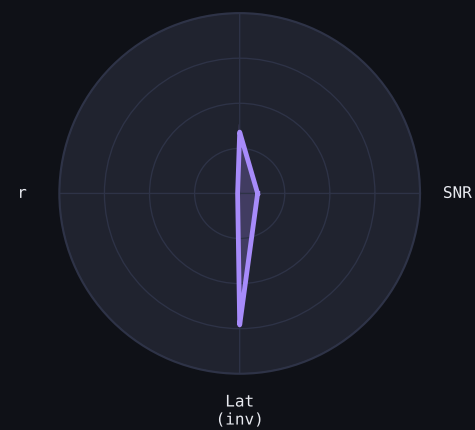
Dual-ch (DE
RMSE
(inv)



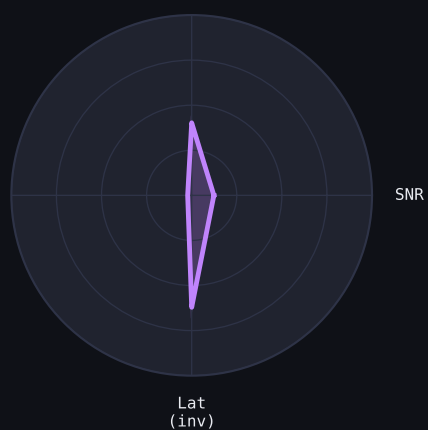
GS0
RMSE
(inv)



LMS
RMSE
(inv)



ϵ -NLMS
RMSE
(inv)



RLS
RMSE
(inv)



CEEMDAN
RMSE
(inv)

